ASSESSMENT 2: CONSTRAINT-BASED PROBLEM SOLVING SOLUTIONS

Course Code: CSA17 Course Name: Artificial Intelligence

1. Doctor Shift Allocation via Backtracking Search

Variables: D1, D2, D3

Domains: {Morning, Afternoon, Night}

Constraints: 1) D1 != Night, 2) D2 must work before D3 (Morning < Afternoon < Night), 3) All doctors assigned unique shifts, 4) D3 != Morning.

Step 1: Assign D1 = Morning (Valid: D1 != Night).

Step 2: Assign D2 = Afternoon (Valid: leaves remaining shifts for D3).

Step 3: Assign D3 = Night (Valid: D3 != Morning, unique shifts, D2 index < D3 index).

Final Valid Schedule: D1 -> Morning, D2 -> Afternoon, D3 -> Night.

2. Robot Grid Navigation with Manhattan Distance (BFS Search)

Problem Constraints: 5x5 grid, uniform move cost = 1, obstacle avoidance, goal-directed navigation.

BFS Execution Details:

Breadth-First Search (BFS) explores nodes level by level using a FIFO queue. Since edge costs are uniform (cost = 1), BFS guarantees finding the shortest path optimal solution.

At each step, the robot generates valid neighbor states (Up, Down, Left, Right) while omitting blocked obstacle cells and grid boundaries.

Node expansion tracks parent pointers to reconstruct the exact optimal coordinate path from Start (S) to Goal (G) once G enters the queue expansion phase.

3. Autonomous Rescue Robot in Disaster Area

a) Constraint Analysis: Limited sensing requires online updating of the local map; obstacle constraints remove invalid state transitions; risk zone costs (+2) modify standard path costs, forcing the search algorithm to balance distance against safety penalties.

b) Solution Approach: Online Search Agent / Uniform Cost Search (A*-like heuristic search). The robot senses adjacent cells, updates its internal graph representation dynamically, computes path costs incorporating both movement and risk weights, and steps forward iteratively.

c) AI Concept Mapping: Intelligent Agent (acts autonomously by perceiving hazards and executing actions); Rationality (maximizes performance with minimum cost and zero safety violations); Environment Type (partially observable, stochastic, sequential, dynamic, discrete).

d) Justification: An online search strategy is mandatory because the disaster building is partially observable; the robot cannot plan a complete offline path without discovering blocked paths dynamically during traversal.